

COURTSIDE: Calling the Point Before It Lands

Predictive Ball Tracking and the Value of Speed in In-Play Tennis Markets

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Abstract. We built a predictive ball-tracking engine that calls a point (ball out, or into the net) before the ball lands. On held-out real table-tennis video our live causal engine made its calls a median 162 ms before the ball reached the table end, none wrong but few (4 of 41 misses), in real time at 119.9 fps; a frame becomes a ready order in 54 ms. On real broadcast tennis (one TV camera, a pretrained ball detector, held-out games) its out calls were right 5 of 5 at the bounce and 3 of 3 when 33 ms ahead. Speed is worth money because, in real Polymarket data on 13,084 ATP and WTA matches (1,310 at Grand Slams), whoever learns the point first sets the price: wallets trading within 3 s of a point earn after fees in all 11 months, and each second of feed delay costs a simulated computer-vision (CV) trader \$42–74 a day. Copying those wallets at their own fills has a Sharpe ratio of 14.5 in sample and 6.7 out of sample, which doubled fees erase. At an assumed 1 s feed our CV trader makes +\$4 a day out of sample pre-registered and +\$57 (Sharpe 8.8) post hoc. Every blind test of a book we could trade failed.

JEL: G14, G12, C58. **Keywords:** computer vision; ball tracking; prediction markets; latency arbitrage; in-play betting; tennis.

1 Introduction and summary

A camera films the court; our model finds the ball in every frame, fits its flight, predicts where it will land and calls the point before the bounce. We ask what that head start is worth: a Polymarket tennis price *is* the chance that a player wins, and quotes lag each point by about a second, so whoever learns it first sets the price. We contribute:

- **Predictive ball tracking**, causal and real time (119.9 fps, 4.6 ms a frame on one GPU), tested on real held-out table tennis and broadcast tennis (Section 4).
- **A wallet-level map of who is fast**: the wallets that trade first earn in all 11 months and everyone else loses; copying them at their fills (v2) has a Sharpe ratio of 14.5 IS, 6.7 OOS.
- **A dollar price for each second of feed delay** from the latency identity of Eq. 1: \$42–74 a day for the first second. With a 1 s feed our order executes 2,119 ms after the point, inside the 3 s bar. Every blind test of a book we could trade failed.

2 Economic hypothesis

A Markov model of points, games and sets gives the exact chance that each player wins from any score (Klaassen and Magnus, 2001), and a Polymarket price is that chance in cents, so each point

Rerun everything with bash reproduce.sh; this paper: python scripts/build_paper.py (<https://github.com/OjasMishra32/computervisionpredictivetabletennismoneymachineouuushiii>). Prices, fills, fees and venue delays are real Polymarket records; only the CV trader's call timing, at an assumed feed latency, is simulated. Paper trading only: no real money, no licensed feed, no court camera.

Real data behind every result; simulated: only CV call timing and the Hawk-Eye-class model

Source	What is real	Matches	Period	Used for
Polymarket books and trades (Polymarket, 2026a)	prices, fills, fees, delays	13,084 (8,554 ATP, 4,530 WTA); 1,310 at Slams	Oct 2025 – Oct 2026	fast tier; v2 (traded in 7,639 IS, 2,003 OOS); CV trader
held back	same	11,307 (10,855 ITF)	same	blind test
Match Charting Project (Sackmann, 2026)	901,233 charted points	5,889 (3,337 men, 2,552 women)	2020–26	how points end (out 36–41%, net 25–26%)
TrackNet broadcast set (Huang et al., 2019)	19,835 labelled frames	10 (95 clips)		court, landing and out-call models
OpenTTGames (Voeikov et al., 2020); Pexels	120 fps video; phone video	41 held-out misses; one rally		early-call engine; tracker end to end

moves fair value by a known amount at a known moment: the *swing* $\Delta(k)$, the match-win chance if point k is won minus that if it is lost, averages 5.7% and reaches 38.4% on a big point (Appendix A).

The news reaches traders in tiers: the court first, then video, then public score feeds 28.2–44.1 s after the price has moved (Fig. A5). A maker whose quote still rests after the point is picked off by whoever knows first: the adverse selection of [Glosten and Milgrom \(1985\)](#) and the latency race of [Budish et al. \(2015\)](#) and [Aquilina et al. \(2022\)](#); in-play betting prices react late, and courtsiders trade on it ([Croxson and Reade, 2014](#); [Brown, 2012](#)). We see them: on 482 points recorded live, a median \$222 of quotes still rested at stale prices 0.25 s before the reprice, and \$0 half a second after it. Polymarket holds each *taker* order (one that hits a resting quote) for 1 s so makers can cancel (a venue setting). The edge is shrinking: the fast tier grew from 4 to 131 wallets as its edge fell 0.20¢ a month ($t = -3.98$); at that pace v2’s OOS +0.60¢ reaches our halving trigger in 1.5 months and zero in 3.0.

H6 (pre-registered). We expect wallets that trade within 3 s of a score move in Polymarket ATP and WTA match markets to earn a positive net *markout* over 30 s (profit at the price 30 s later, after fees), because the other side is a maker who has not yet cancelled a stale quote. The edge persists because physics and data licences decide who learns first. If so, fast wallets picked on past months stay positive every month and out of sample; H6 fails if the markout is ≤ 0 in over a third of months, or out of sample. H1–H4 were committed before any result (7232986, 2026-10-03 09:50 UTC); H6 was written after the in-sample tier results (D5) and frozen before we opened the held-out data.

3 Data and universe

Of the table’s 1,310 Grand Slam matches, 415 are at the Australian Open, 412 at Roland Garros and 483 at the US Open (928 main draw); Wimbledon is a separate Polymarket series outside our universe. The CV trader works in the 7,755 matches from May 15 on, when the hold was 1 s (895 of them at Slams), trading in about 681 IS and 246 OOS per seed. The universe: every resolved ATP/WTAs singles match market with at least \$5,000 traded (\$2.84B in all). *In sample* (IS) is the first 80% of matches by start time; *out of sample* (OOS) is the last 2,617, from 2026-08-25 14:15 UTC. We looked at the OOS while building v2 and the CV simulation: it is *burned*. For a blind test, which also checks survivorship, we held back 11,307 smaller markets and opened them once per frozen rule. Six tapes hit the API’s offset cap, 2,889 early matches paid no fee, 2.89% of markets settled 50/50 (kept), tape timestamps run 1.98 s late, no corporate actions apply. Timing: ESPN and the WTA ([ESPN, 2026](#); [WTA, 2026](#)).

4 Methodology

The fast tier and v2. We flag a score move when the 10 s volume-weighted price moves at least 4¢ away from its prior 60 s. Each month a wallet joins the fast tier if, in earlier months, it traded at least 30 times 0–3 s after a flag, in at least 10 matches, with $t > 3$. v2 copies their trades, sizes each by its

resolution risk (the best of 55 sizing rules, picked walk-forward), caps orders at \$1,000 and exposure at 100 shares net per match, and holds to resolution. At their own prices it measures what their speed is worth, not what we could earn: copying the same trades 3 s later loses (Fig. 1a).

Being fast ourselves. Our second strategy: a computer-vision (CV) model watches the video, calls the point early and sends a taker order before the stale quote is cancelled. We have no video of the matches we trade, so we simulate this trader on past points with an assumed feed latency (licensed feed not purchased); all other parameters are measured. The order wins only if

$$V + \delta + D - \ell < L + R \iff V < L + R - D - \delta + \ell, \quad (1)$$

where V is the feed delay (how late the video reaches us), δ our pipeline plus network time, D the 1 s taker hold, ℓ how early the CV calls the point, L the *stamp lag* (how long after the ball lands the official point stamp is logged) and R when the book reprices relative to that stamp. On 482 points the book moved 1.2 s before the stamp. Nobody publishes L , so we show two readings: 2.0 s, which we *pre-registered* (written before any result; PREREG.md, 17:25 UTC), and 3.14 s, a *post hoc* estimate from one day of data. The simulated trades are past points where the price later moved $\geq 4\text{¢}$: *selected on outcomes, not ex ante*.

Our vision on real video. On the TrackNet broadcast set we built court calibration, a 3D flight fit and landing predictors, and tested them once on games 8–10 (132 bounces). From one TV camera, on the pretrained detector’s own track, the landing point is 17 cm off at the bounce and 0.7–1.2 m off when predicted 33–300 ms ahead (median). With thresholds set on games 1–7, out calls were right 5 of 5 at the bounce (half the outs), 3 of 3 at 33 ms and 3 of 4 at 67 ms ahead, and fail from 100 ms (2 of 5). Only the bounce-time call met our 95% precision rule on games 1–7, and the detector weights (yastrebk, 2026), not ours, saw frames of all ten games, so these are in-distribution numbers. On simulated Hawk-Eye-class physics our spin-aware model cuts the landing spread 10-fold at 200 ms (out-call precision 1, recall 0.98). On held-out table tennis at 120 fps our live causal engine called 4 of 41 misses early (median lead 162 ms), none wrongly, at 119.9 fps on one L4 GPU; outside play it makes about 30 phantom calls an hour, and a rally gate (trade a miss call only within 2.0 s of a bounce call, set before the test) removes 5 of 7 calls on unlabelled balls in its held-out log but keeps only 3 of 5 correct calls and leaves about 8 an hour: not yet safe to trade. With a simulated 1 s feed, our 54 ms pipeline, 65 ms of network and the 1 s hold, an order can execute 2,119 ms after the point (p99 2,218 ms; 24 runs, order built, not sent), inside the organizers’ 3 s bar (Fig. A1).

Costs and validation. Takers pay $f(q) = r q (1 - q)$ a share at price q , or $10,000 r (1 - q)$ bps of notional; today $r = 5\%$, so a trade at 50¢ pays 1.25¢. Each match pays its own fee and every trade the spread it crossed; we also double both. v2 paid 120 and 179 bps in fees (IS, OOS) and still netted 273 and 114 bps. We tune only in sample, walk-forward (fit on past months, test on the next).

5 Results

The copy book. The fast tier earns every month and everyone else loses (Fig. 1a). v2 earns in both periods (Table 1) but thinly: entering half a tick worse leaves +0.10¢ a share OOS, and doubled fees turn it to −0.34¢ (Fig. 1b). It stays positive in every fee and hold regime, +1.96¢ to +1.02¢ (Fig. A3).

A Sharpe ratio of 14.5 is a red flag. We assumed a bug and found one: the detector’s start time used hindsight (deviation D9); fixing it cut the in-sample Sharpe from 16.8 to 14.5. The rest comes from about 270 small, nearly independent bets a day (mean same-day correlation 0.0013). The *deflated Sharpe* (Bailey and López de Prado, 2014), the chance the Sharpe is real once every variant is counted, is 0.997 in sample but 0.075 out of sample: forty days cannot rule out luck. Nearby settings work too: all 55 sizing rules have a per-share 95% CI above zero in sample (Sharpe 4.1–16.8), the v2-safe grid’s probability of backtest overfitting is 9–24% across block choices (A6), and across all 14 feed delays from 0 to 60 s the CV profit never rises by more than \$0.09 a day (Fig. A2).

What speed is worth. Pre-registered, the CV trader breaks even at a 1.01 s feed delay out of sample (Table 2; curves in Fig. A2): the call must reach the venue 0.89 s before the umpire’s stamp. Post hoc it

Table 1. *Our copy of the fast tier's trades (v2), at their prices: the edge holds out of sample but is thin.*

Return	IS	OOS	Risk and costs	IS	OOS
Net ¢ a share	+1.38	+0.60	Max drawdown	-2.0%	-2.1%
95% CI	[1.17, 1.59]	[0.09, 1.13]	Worst month	+\$2,543	-\$434
Sharpe ratio	14.5	6.7	Skew	+0.53	+0.33
95% CI	[11.9, 17.4]	[1.9, 12.3]	Turnover, × a year	93	129
Deflated Sharpe	0.997	0.075	Months positive	7/7	2/3
Annual return	253%	148%	Net ¢, fees ×2	+0.77	-0.34
Annual volatility	17.5%	22.2%	Net ¢, all costs ×2	+0.27	-0.84

Notes: 55,662 trades in 206 days IS (+\$40,426), 10,412 in 40 OOS (+\$3,688; burned, not blind), net of costs. Sharpe: daily P&L $\times \sqrt{365}$, block-bootstrap CI; ¢ CI: bootstrap over matches (over wallets, OOS: [-0.45, 2.38]); deflated for 3,386 trials (Appendix A).

a Only the fast tier earns; copying late loses **b** v2 earns; doubled costs erase the OOS gain

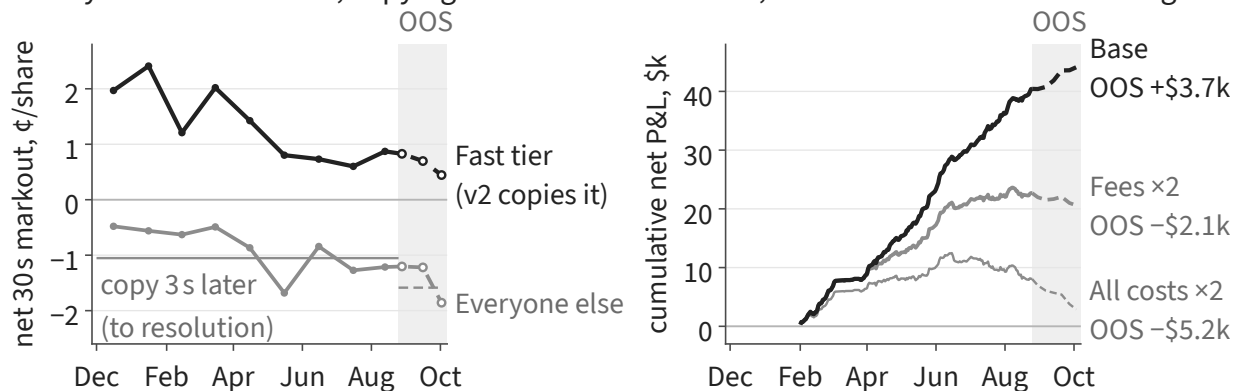


Figure 1. *Only the fast tier earns, and the edge is thin.* (a) Monthly net 30 s markout (copying 3 s later: held to resolution). (b) v2's net P&L, base and doubled costs. Source: results/v2.

makes +\$57 a day at 1 s (+\$8 at the low end of the stamp-lag interval), but read per point the post hoc inference loses (-\$17 a day). Every reading loses at 3 s, so meeting the 3 s bar is not enough. A replay of 9 matches recorded live on Oct 3 against their real books, calling every point ex ante, loses in 36 of 36 settings (-0.94¢ a share at 1 s; Fig. A4): the gap is knowing which points matter in advance.

Blind tests. On 11,307 markets we had never examined (10,855 ITF; pre-registered at 337bf7c before fetching any tape), frozen v2 earned +2.02¢ [1.14, 2.93] a share IS but +1.22¢ [-0.19, 2.65] OOS, so it fails. Pre-registered secondary statistics show the mechanism holding blind: the fast tier beat the rest by +3.09¢ [2.89, 3.28] IS and +2.16¢ [1.79, 2.53] OOS, and v2's 30 s markout was +0.59¢ [0.13, 1.03] OOS. The maker book and CV rule v3 also failed (Appendix B). Clean: H6, the blind test and the vision tests; burned: v2 (fit after v1's single OOS read, shown on the OOS twice; D8, D10) and the CV OOS. H6 and Fig. 1a are an onset-aligned event study; tradeable numbers use the causal window (D9). Over 4,219 variants and 85 held-out reads we changed rules after a look 2 times for v2 and 13 for the CV simulation, and chose the post hoc reading and 1 s base case after looking. Blind forward test: pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5).

6 Risk management

Limits and de-risking. Each order is at most \$1,000 and each match at most 100 shares net at 5–95¢, which caps the open position at \$95, not the loss: buying both sides of a match can lock in more (worst match -\$206 IS, -\$154 OOS), so a \$1,000 daily loss ends the day. Set in advance, we stop if the

Table 2. *The CV strategy at three assumed feed delays: pre-registered it breaks even near 1 s.*

Feed delay V	In sample			Out of sample (burned)		
	\$ a day	Sharpe	¢ a share [95% CI]	\$ a day	Sharpe	¢ a share [95% CI]
A. Pre-registered stamp lag, $L = 2.0$ s; break-even V: 1.09 s IS, 1.01 s OOS						
0.5 s, best case	+\$28	4.0	+0.61 [0.08, 1.14]	+\$15	2.1	-0.02 [-1.22, 1.09]
1 s, base case	+\$15	2.1	+0.40 [-0.32, 1.08]	+\$4	0.3	-0.38 [-2.08, 1.21]
3 s, the organizers' bar	-\$13	-2.6	-1.35 [-3.15, 0.37]	-\$13	-2.6	-1.69 [-4.85, 1.32]
B. Post hoc estimate, $L = 3.14$ s (95% CI 2.23–3.22 s); break-even V: 2.23 s IS, 2.14 s OOS						
0.5 s, best case	+\$133	15.5	+1.21 [1.00, 1.42]	+\$81	12.2	+0.82 [0.39, 1.25]
1 s, base case	+\$94	11.9	+1.11 [0.85, 1.38]	+\$57	8.8	+0.66 [0.10, 1.19]
3 s, the organizers' bar	-\$6	-1.4	-0.79 [-2.26, 0.60]	-\$11	-2.4	-1.47 [-4.45, 1.34]

Notes: simulated, assumed feed latency (licensed feed not purchased); parameters measured; 20 seeds a cell. Trades: past points where the price later moved $\geq 4\text{¢}$, selected on outcomes, not ex ante. Call rates are the offline rule's; with our causal engine's own calls the 1 s cells are +\$17 / +\$4 (pre-registered) and +\$98 / +\$57, Sharpe 8.8 OOS (post hoc), IS / OOS, so no look-ahead in the calls drives the table. Per-tournament timing came after the first P&L (deviation V3).

score feed is over 2 s stale, vision over 1 s stale, or latency above its 95th percentile; we halve size if the trailing 30-day edge falls below 0.3¢, and stop if it reaches zero, if drawdown hits 5%, or if the venue changes its fee or hold. None fired in sample (trailing edge $\geq +0.74\text{¢}$, worst day $-\$551$).

Model, concentration and tail risk. Wrong CV calls always fill, so an out call trades only at confidence ≥ 0.95 and one confirmed false call halts the day; 2.5 phantom calls a day would erase the pre-registered 1 s OOS P&L (32 post hoc). The top five copied wallets carry 81% of v2's P&L in sample and 142% out of sample; without them v2 earns -0.34¢ [-1.20, 0.46] OOS. A per-wallet cap (\$1,000 a day, and a wallet whose trailing edge turns negative is retired), chosen in sample from 12 more variants and pre-registered at 4a5a19d, ran once on the burned OOS (non-blind): $+0.91\text{¢}$ [0.20, 1.66], Sharpe 7.6, worst day $-\$284$ instead of $-\$469$; it costs 37% of in-sample dollars and the top five still carry 129%. The UMA oracle settles markets (Polymarket, 2026c), US persons may only close positions (Polymarket, 2026b), and courtsiding breaches ticket terms (International Tennis Federation, 2026). Factor risk is nil (largest factor $|t| = 1.36$; α has $t = 8.48$) (Fama and French, 1993; Carhart, 1997).

7 Liquidity and capacity

How much capital. v2: \$22,754, the largest size (1×) at which its OOS CI stays above zero (5× loses); it trades \$7,182 a day, 0.07% of match volume. The CV book: \$40,000–\$73,000 post hoc (OOS-IS), where its Sharpe halves, and at most \$177 a day (on \$83,000 OOS; Fig. A6); pre-registered, none (Sharpe 2.7 IS, 0.8 OOS at its smallest size). It takes at most half the stale depth at each point: 0.22% of in-play volume but 9% of what the fast tier trades 0–3 s after a point. These are upper bounds: we model no price impact beyond the stale depth, and up to 131 fast wallets compete for the same quotes. At 1 s the book could pay at most \$1,644 a month for data post hoc and \$55 pre-registered, against feed prices of \$1,250–\$10,000. At today's fee COURTSIDE prices speed; it is not yet a business.

8 Limitations and next steps

The stamp lag L is unmeasured and moves the break-even one for one (Eq. 1); the 1 s hold dominates it, since even for a 20 ms bot the inferred lag is 2.85 s. One session with a licensed feed and a few test orders would measure L ; at 1 s the feed pays for itself only if $L \geq 3.0$ s, a purchase we have not made. Our out calls are tested on real broadcast tennis and table tennis, never on the matches we trade; trading is simulated and we never sent an order. A fee rise, a shorter hold or more fast wallets would break it.

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Disclosure. We used no real money and connected no funded account. Simulated parts are labelled where they appear. We hold no licensed data feed, no footage of the matches we trade and no court access. Public datasets we did not build: the Match Charting Project (CC BY-NC-SA 4.0), the TrackNet broadcast tennis set and the OpenTTGames dataset. Open-source parts we did not build: the BlurBall and WASB model weights (Gossard et al., 2026; Tarashima et al., 2023), and the TrackNet weights and TennisProject/TennisCourtDetector code (yastrebkvs, 2026) (used unmodified, for evaluation only). The tennis rally is a Pexels stock clip (Gelato Prod, 2021); the logo on page 1 is the Gator Quant Hacks 2026 event's own mark. Authors: Ojasva Mishra, Yoan Exposito, Rafael Penhas, Ian Hoang (University of Florida).

Appendix A Calculations

Everything needed for scoring is in the main pages. Below: formulas with worked examples (A), every test (B), extra figures (C) and the records (D). Each value is in results/paper/numbers.json with its source file.

A1. Sharpe ratio. $SR = \sqrt{365} \bar{d} / s_d$, where d_t is net P&L on every calendar day (zero on days without trades) and s_d its standard deviation. v2 IS: $\bar{d} = \$196.24$, $s_d = \$258.87$ over $T = 206$ days, so the daily ratio is 0.758 and $SR = 0.758 \times 19.105 = 14.48$. OOS: $\$92.19 / \$264.15 \times 19.105 = 6.67$.

A2. Sortino ratio. $\sqrt{365} \bar{d} / DD$, where the downside deviation is $DD = \sqrt{\frac{1}{T} \sum_t \min(d_t, 0)^2}$. For v2 OOS, $DD = \$125.64$, so the ratio is $\sqrt{365} \times \$92.19 / \$125.64 = 14.0$.

A3. Maximum drawdown. $MDD = \min_t (E_t - \max_{s \leq t} E_s)$, where E_t is cumulative P&L, as a share of capital. v2 OOS: $-\$469 / \$22,754 = -2.06\%$.

A4. Probabilistic Sharpe. The chance that the true daily Sharpe beats SR^* , given the sample length T , skew γ_3 and kurtosis γ_4 (Bailey and López de Prado, 2014):

$$PSR(SR^*) = \Phi\left((\hat{SR} - SR^*)\sqrt{T-1} / \sqrt{1 - \gamma_3 \hat{SR} + \frac{\gamma_4 - 1}{4} \hat{SR}^2}\right).$$

v2 OOS: $\hat{SR} = 0.349$, $T = 40$, $\gamma_3 = 0.326$, $\gamma_4 = 3.26$, so $PSR(0) = 0.987$.

A5. Deflated Sharpe. $DSR = PSR(SR_0)$, where SR_0 is the best Sharpe we would expect from N tries on pure noise, with γ the Euler–Mascheroni constant:

$$SR_0 = \sqrt{V} \left((1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{Ne}\right) \right).$$

v2 OOS with $N = 3,386$ and $V = 1/(T-1)$: the two Φ^{-1} terms are 3.436 and 3.698, $SR_0 = 0.574$ a day (10.97 a year), and $DSR = 0.075$. Table 1 shows the lowest DSR over three choices of V .

A6. Probability of backtest overfitting. (Bailey et al., 2017). Split the daily P&L of 24 variants into $S = 16$ blocks. For each of the $\binom{16}{8} = 12,870$ ways to call half the blocks in sample, take the best in-sample variant, find its rank ω out of sample and set $\lambda = \ln \frac{\omega}{1-\omega}$; $PBO = \Pr(\lambda \leq 0)$. v2-safe grid: $PBO = 15\%$ (median $\lambda = 1.27$).

A7. Stationary block bootstrap. (Politis and Romano, 1994). Resample days in blocks of random length (mean 5 days) 10,000 times and read the percentiles. v2 OOS Sharpe: [1.9, 12.3], standard error 2.65, $P(SR \leq 0) = 0.0023$.

A8. Factor regression. $r_t = \alpha + \beta_M \text{MktRF}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_U \text{Mom}_t + \varepsilon_t$ on daily return on capital, with Newey–West errors (5 lags) (Fama and French, 1993; Carhart, 1997; Newey and West, 1987; French, 2026). v2 IS (138 days): $\alpha = 0.801\%$ a day ($t = 8.48$); $\beta_M = 0.173$ ($t = 1.36$), $\beta_S = 0.085$, $\beta_H = 0.044$, $\beta_U = 0.028$; $R^2 = 3.3\%$.

A9. Taker fee. $f(q) = r q(1 - q)$ a share at price q , or $r(1 - q)$ per dollar, which is 10,000 $r(1 - q)$ bps. At $r = 5\%$: $q = 0.5$ pays 1.25¢ (250 bps) and $q = 0.8$ pays 0.80¢ (100 bps).

A10. Point swing. $\Delta(k)$ of Section 2, with the match-win probability computed exactly by recursion over points, games, tiebreaks and sets (src/markov.py). Two equal players who win 0.645 of service points: at the first point $\Pr(\text{win}) = 50.0\%$, moving to 51.2% or 47.8%, so $\Delta = 3.4\%$. Serving at 5–5, 30–40 in the deciding set, $\Pr(\text{win}) = 33.7\%$, moving to 47.3% or 8.9%, so $\Delta = 38.4\%$. The mean $|\Delta|$ over a match is 5.7%.

A11. Latency budget. Eq. 1. With the pre-registered $L = 2.0$ s, $D = 1$ s and our measured $\delta = 0.119$ s, a point whose book reprices at the median $R = -1.16$ s needs $V < -0.28 + \ell$: no feed wins it without an early call. At the 90th percentile, $R = 0.08$ s and $V < 0.96$ s. The sweep's break-even (1.09 s IS) comes from these late reprices.

A12. Capital and return on capital. Capital is three times the peak dollars locked in open positions, each held 4 h or until resolution. v2 IS: peak \$9,434, capital \$28,302; return = $365 \bar{d} / \text{capital} = 365 \times \$196.24 / \$28,302 = 253\%$ a year.

A13. Capacity and volume share. Capacity is the capital at which the seed-mean Sharpe falls to half its smallest-size value; share is our notional a day over market volume a day. CV, post hoc, IS, just below half Sharpe: \$10,428 a day against \$4.71M of in-play volume (0.22%) and \$117,047 traded by the fast tier 0–3 s after a point (9%).

Appendix B Everything we tested

Table A1. Every test we ran, its best result and our verdict. IS: in sample; OOS: out of sample (burned unless marked blind); ¢: net cents a share.

Test	What it was	Best result	Verdict
<i>Who is fast, and what they earn (public tapes)</i>			
H1 follow the jump	take the direction of a ≥ 4 ¢ move, hold 30 s	OOS -2.08 ¢ $[-2.17, -1.98]$; 20 of 20 IS variants below zero	rejected
H2 favourites	buy in-play favourites priced 85–97¢	OOS $+0.80$ ¢ $[-1.36, 2.66]$	no edge
H4 score feeds	do public score feeds lag the book?	median lag 44.5 s ($n = 75$)	confirmed
H5 quote after jumps	post maker quotes after a move	OOS $+0.16$ ¢ $[-0.49, 0.87]$	not significant
H6 fast tier	wallets trading 0–3 s after a move, picked on past months	$+0.93$ / $+0.75$ ¢ net 30 s markout (IS / OOS); 9/9 and 3/3 months positive	confirmed
Kalshi lead-lag	Kalshi vs Polymarket on the same points	Kalshi first in 68.9% of 106,347 repricings (median 5 s)	lag mostly mechanical
Signal decay (TT5)	with-jump markout at 1–2 s minus 10–30 s after a move	OOS $+0.66$ $[0.45, 0.87]$ ¢; whole year $+0.19$ $[-0.26, 0.46]$ ¢; fast tier $+1.18$ to $+0.59$ ¢ (IS, 0 to 1–2 s)	decays in today's regime only
Clocks	when the book, the stamp and the tape move	book 1.2 s before the stamp ($n = 482$); tape 1.98 s late	measured
Factor regression	Fama–French three factors and momentum	$\alpha t = 8.48$; largest factor $ t = 1.36$	no factor exposure
<i>Copying the fast tier (v2 and relatives)</i>			
v1	copy the fast tier, dollar sizing; OOS opened once	IS $+1.16$ ¢; OOS $+0.57$ ¢, $-\$36,056$	failed

Test	What it was	Best result	Verdict
Sizing	55 sizing policies, walk-forward	G_50pct_net100: Sharpe 14.5 IS; all 55 with IS CI above zero (Sharpe 4.1–16.8)	adopted (v2); plateau
Exits	maker exit instead of holding (902 variants)	IS +0.53¢ [0.42, 0.64] (1 s / 5% regime); live fills adverse (27% filled in 30 s)	not adopted
Trade selection	which fast-tier trades to take (341 variants)	nested walk-forward +1.76¢ [1.36, 2.16] vs +1.16¢ (IS)	IS only
Side markets	side markets vs the match market (68 variants)	leaning maker +2.73¢ [1.73, 3.70] (IS); taking stale side quotes: no edge	IS only; tiny
v2	frozen copy book at the fast tier's own prices	Sharpe 14.5 / 6.7; OOS +0.60¢ [0.09, 1.13]	headline (Table 1)
v2, worse entry	enter ½ or 1 tick worse	IS +0.88 / +0.38¢; OOS +0.10 / -0.40¢	about zero
v2, costs doubled	fees ×2 / all costs ×2	IS +0.77 / +0.27¢; OOS -0.34 / -0.84¢	fails OOS
v2, fixed costs	after central data and running costs	IS +\$29 a day; OOS -\$75 a day	fails OOS
v2, unseen markets	11,307 markets we never examined (blind)	IS +2.02¢; OOS +1.22¢ [-0.19, 2.65]	fail
v2-safe	net cap 50, fixed after v2's OOS losses	Sharpe 15.7 / 9.3; blind +1.21¢ [-0.02, 2.43]	fail (blind)
Per-wallet cap	\$1,000 a wallet a day plus retirement, chosen on IS from 12 variants, pre-registered	OOS +0.91¢ [0.20, 1.66], Sharpe 7.6; top five 129% (was 142%)	pass (burned OOS)
Maker v1	pre-registered maker book (blind)	+1.87¢ [-0.14, 3.86] per fill, -\$379	fail
Rigor pack	deflated Sharpe, PBO, block bootstrap	DSR 0.997 / 0.075; PBO 15% (9–24% by block choice); $P(SR \leq 0) = 0.0023$	luck not ruled out OOS
Blind forward test	v2 on new matches, CV rule v3 as secondary	pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5)	not run
<i>Being fast ourselves (computer vision)</i>			
H3 early call	table-tennis MISS call 50 ms before contact, precision ≥ 0.95	live causal engine: 4 of 41 misses, none wrong; offline evaluation (look-ahead feature): 11/11	precision passes; recall low
Spin model, table tennis	spin-aware early call, held-out footage	10/12 right at 50 ms vs the frozen model's 11/11 (both offline)	not adopted
Tennis tracker, real clip	our tennis tracker on a Pexels rally (no labels)	ball in 89% of 300 frames; 48/48 spot-checks on the ball; court lines 0.36 px (median)	tracks; no in/out calls
Broadcast tennis	single-camera landing and out calls, TrackNet games 8–10 (tuned on 1–7)	out calls 5 of 5 at the bounce, 3 of 3 at 33 ms, 3 of 4 at 67 ms; landing 17 cm at the bounce, 0.7–1.2 m at 33–300 ms	bounce only at 95%; detector saw all games
Tennis spin model	spin-aware landing model (simulated physics)	0.57 vs 5.76 cm at 200 ms; OUT precision 1, recall 0.98	simulation only

Test	What it was	Best result	Verdict
Rally gate	a miss call trades only within 2.0 s of a bounce call (engine log; 0.5–1.2 s also run)	removes 5 of 7 calls on unlabelled balls; keeps 3 of 5 right ones (2 at 0.5–0.8 s)	not enough
GPU engine	streamed causal engine on one L4	119.9 fps; 4.6 ms p50; 0 frames dropped	measured
WebRTC pipeline	our clip over WebRTC into the engine	46 ms p50, 67 ms p99 frame to call; video leg 4–8 ms	measured
End to end	video frame to unsigned order on a live book	54 ms ours; 2,119 ms with a 1 s feed vs 3,000 ms	pass (paper)
Courtside camera	CV trader with a camera at the venue ($V = 0$)	+\$89 / +\$46 a day (pre-registered)	simulated; no camera
Latency sweep	V from 0 to 60 s, six stamp-lag readings (944 cells)	pre-registered break-even 1.01–1.09 s	Table 2
CV rule v3	rule tuned on 360 IS variants	IS -0.22¢ ; blind -0.52¢	fail
Replay, all points	9 matches' real books, every point called	-0.94¢ $[-1.28, -0.58]$ at $V = 1$ s; 36 of 36 settings below zero	loses
Replay, big points	only points with a large Markov swing (exploratory)	-0.95¢ $[-1.64, -0.40]$; 9 of 9 cells below zero	loses
Capacity	size grid for the CV book and v2	post hoc Sharpe halves at \$40,000–\$73,000; pre-registered: none	small
Table-tennis markets	TT1–TT5 on Polymarket table tennis	no fast wallet qualifies; spread 94¢	untestable

Appendix C Robustness and extra figures

These figures back the pipeline, regime, replay, latency and capacity claims in the main text.

Frame to executable order in 2.12 s; our part 54 ms

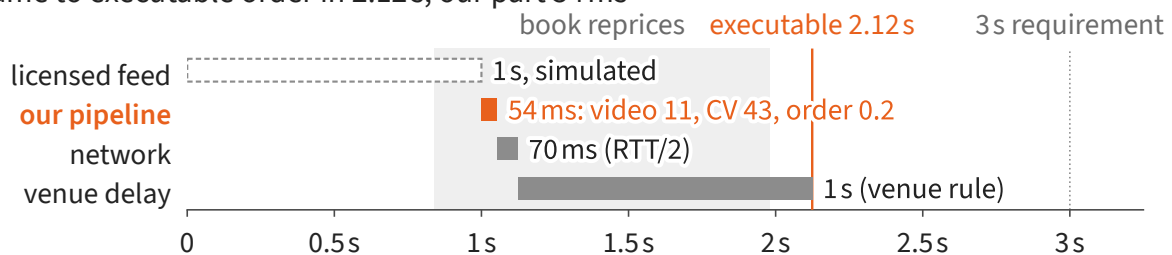


Figure A1. We can trade inside 3 s. Mean stage times over 24 runs on our own footage; the 1 s feed is simulated; order built, not sent. Shaded: when the book reprices. Source: results/e2e.

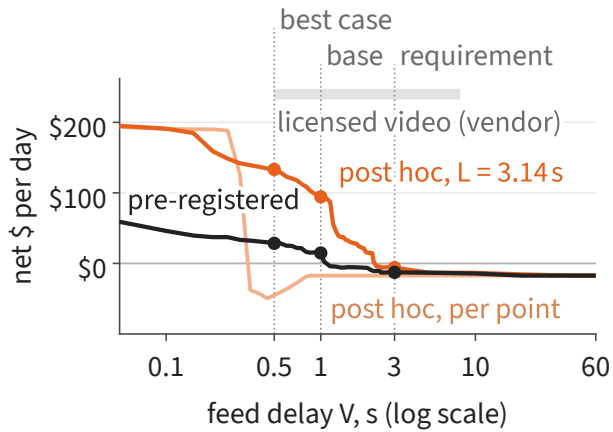
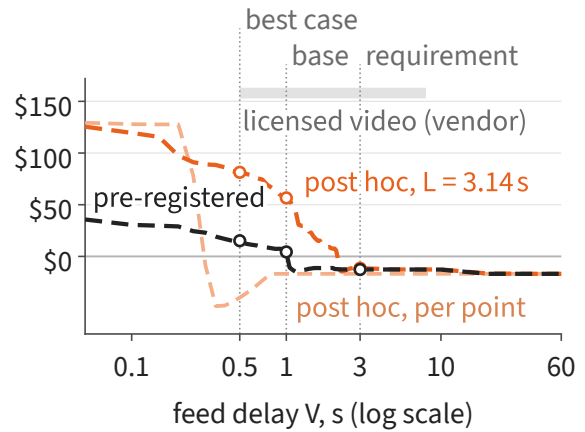
a IS: zero profit at 1.1–2.2 s of delay**b** OOS: zero profit at 1.0–2.1 s of delay

Figure A2. Each second of feed delay costs money. Simulated CV strategy, (a) IS, (b) burned OOS; dots: Table 2; grey bar: vendor-stated licensed video. Assumed feed latency (licensed feed not purchased); parameters measured. Source: results/tier0.

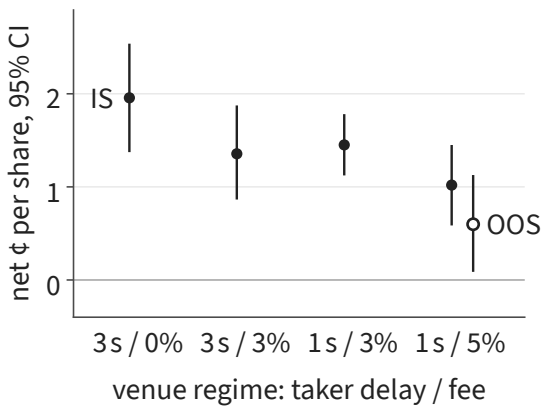
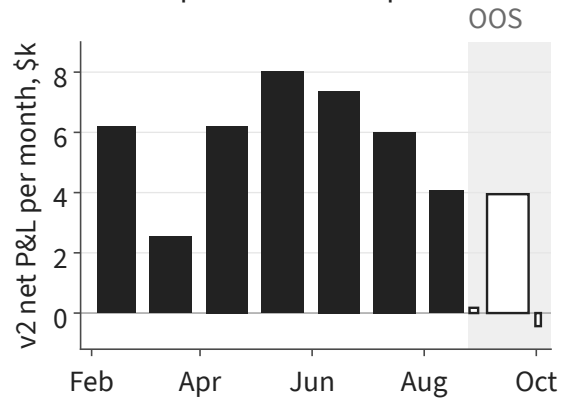
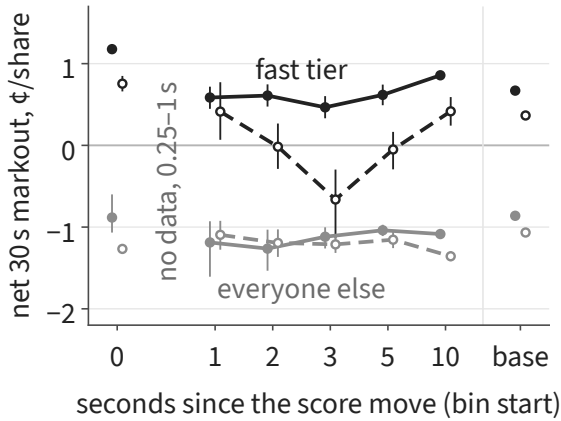
a Edge stays positive in every venue regime**b** 7 of 7 in-sample months are positive

Figure A3. v2's edge shrinks as the venue's protection shrinks, and stays positive in every regime. (a) Net \$ a share by venue regime (taker hold / fee), 95% CIs; (b) net P&L by month, at the fast tier's own prices. Source: results/risk/risk_stats.json.

a The fast tier stays ahead at every horizon



b Trading every point loses at every delay

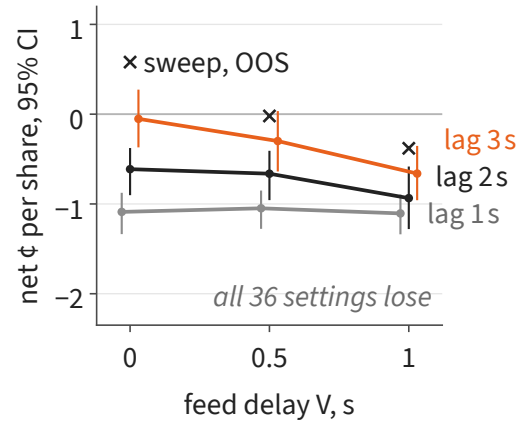
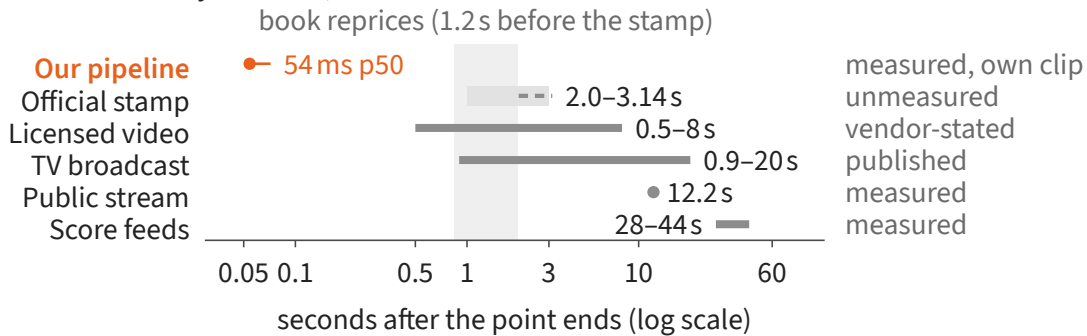


Figure A4. *The fast tier leads at every horizon; trading every point on real books loses.* (a) Net 30 s markout by seconds since the score move; filled: IS, hollow: OOS. (b) Replay of 9 matches recorded live on 2026-10-03 against their real order books, every point called; net ¢ a share marked at +30 s with match-clustered 95% CIs; ×: the sweep's OOS cell. One day, assumed feed latency, paper only. Source: results/{decay,replay}.

a Our order is ready in 54 ms; feeds take seconds



b Live engine: 4 of 41 misses called early, none wrong

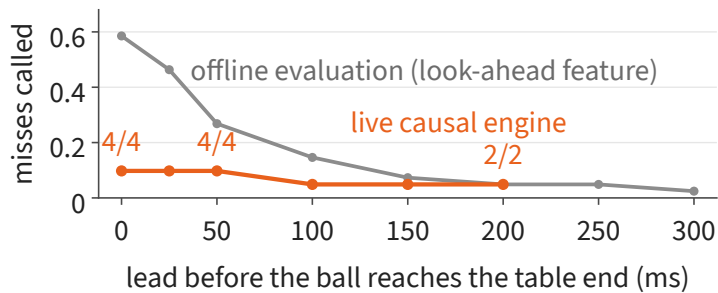
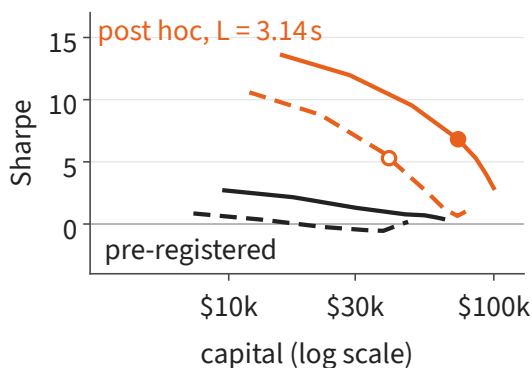


Figure A5. *Who sees the point first.* (a) Delay after the point ends (log scale): our pipeline (dot p50, whisker p99, own clip over WebRTC, laptop); the official stamp, dashed from the pre-registered 2.0s to the post hoc 3.14s over the cited 1–3s band (Regen Sports, 2026; Stats Perform, nd); vendor-stated licensed video (Stats Perform, nd; NEXT.io, nd; Sportradar, nd; Ably, nd); published stream delays (ISPreview, 2021; GL Systemhaus, 2018; The Desk, 2026); measured public stream and score feeds. Shaded: when the book reprises. (b) Early out-calls on 41 held-out table-tennis misses: the live causal engine (orange), which we can deploy, and the offline evaluation (grey), which used a look-ahead feature. Source: results/{e2e,webrtc,tracking,engine}, research/v2/latency.

a Sharpe halves by \$40k–\$73k (post hoc)



b The cheapest data stack needs \$14k–\$16k

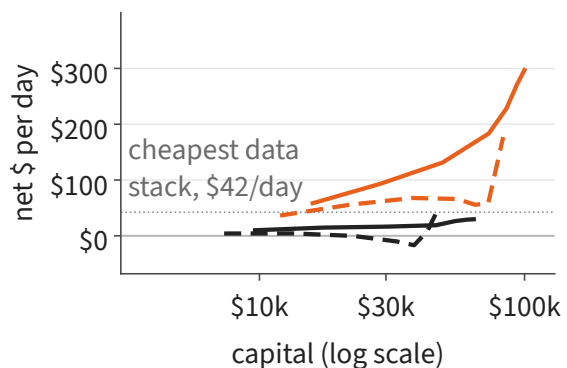
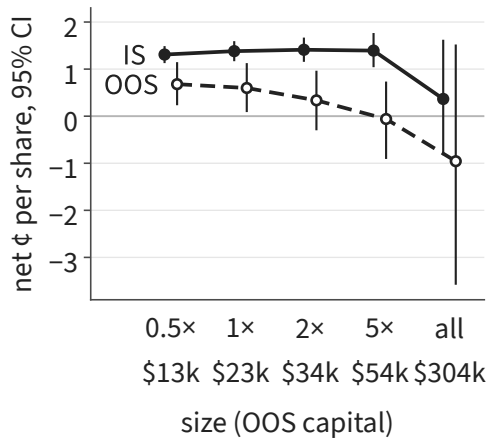


Figure A6. *How much capital the CV strategy can use at a 1 s feed.* (a) Sharpe and (b) net \$ a day against capital (net cap 50 to 5,000 shares, \$250 orders, 10 matches a day, half the stale depth); circles: where the post hoc Sharpe halves; dotted: the cheapest data stack. Solid: IS; dashed: burned OOS. Simulated, assumed feed latency (licensed feed not purchased); parameters measured. Source: results/capacity.

a v2's OOS edge fades as size grows



b After fixed costs only v2 IS stays positive

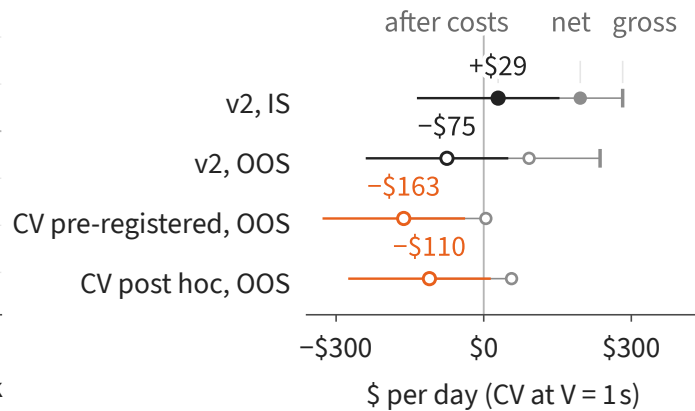


Figure A7. v2's capacity is small and fixed costs bind. (a) Net edge by size (caps scaled; optimistic at large size, since we model no price impact); filled: IS, hollow: OOS. (b) \$ a day gross, after fees and after central fixed costs (bar: low to high); CV rows at the 1s base case, burned OOS, pre-registered first; assumed feed latency (licensed feed not purchased); parameters measured. Source: results/financials.

Appendix D Records

We keep the full logs in the repo instead of printing them here.

Pre-registration. H1-H4: HYPOTHESIS.md, committed at 7232986 (2026-10-03 09:50 UTC) before any result. H6, v2 and the forward test: HYPOTHESIS_V2.md (amendments A1-A5). Table tennis: HYPOTHESIS_TT.md. The CV simulation's research/v2/tier0/PREREG.md says it was written at 17:25 UTC; it was committed with its results at 19:24 UTC (a5769c7), and the peek log shows the OOS grid first ran at 17:29 UTC. git log on these files lists 19 commits.

Deviations. DEVIATIONS.md and research/v2/*/DEVIATIONS*.md give every rule change and its reason: 2 changes after a look for v2 and 13 for the CV simulation (T1-T3, V1-V10).

Reads of held-out data. results/oos_peeks.log has 85 lines, one per read: 7 blind first runs, 18 non-blind reads, 6 descriptive checks, 27 audits or re-runs, 22 live-data or replay reads and 5 others; 2 were written after their run and say so.

Variants. 4,219 in all (results/paper/variants.json): H1-H6 pre-registered and H5/H6 rules (44); v2 lenses (exit, sizing, selection, cross-market, latency, Kalshi) (3,342); v2-safe grid (24); tier-0 pre-registered scenarios (432); tier-0 v3 in-sample grid (360); maker v1 evaluated variants (8); cost-stress cases (4); table-tennis tests TT1-TT5 (5). Sensitivities that chose nothing: 944 latency-sweep cells and 36 replay cells, plus 4 rally-gate values (Section 4). The per-wallet cap of Section 6 added 12 more variants after this count. Two presentation choices were post hoc: the 3.14 s headline reading and the 1 s base case.

Grids and sources. Latency sweep: results/tier0/latency_sweep.json; replay: results/replay/selective/selective.json; fixed costs: results/financials/financials.json; latency sources: research/v2/latency/results.json; every number with its source: results/paper/numbers.json.

Forward test and live session. Blind forward test: pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5). The live paper session was stopped by a team decision and is not used (research/v2/maker/DEVIATIONS_LIVE.md, L15).